

ALP

WORKSHEET
TEACHING ACTIVITY

Algorithmics and Data Structures

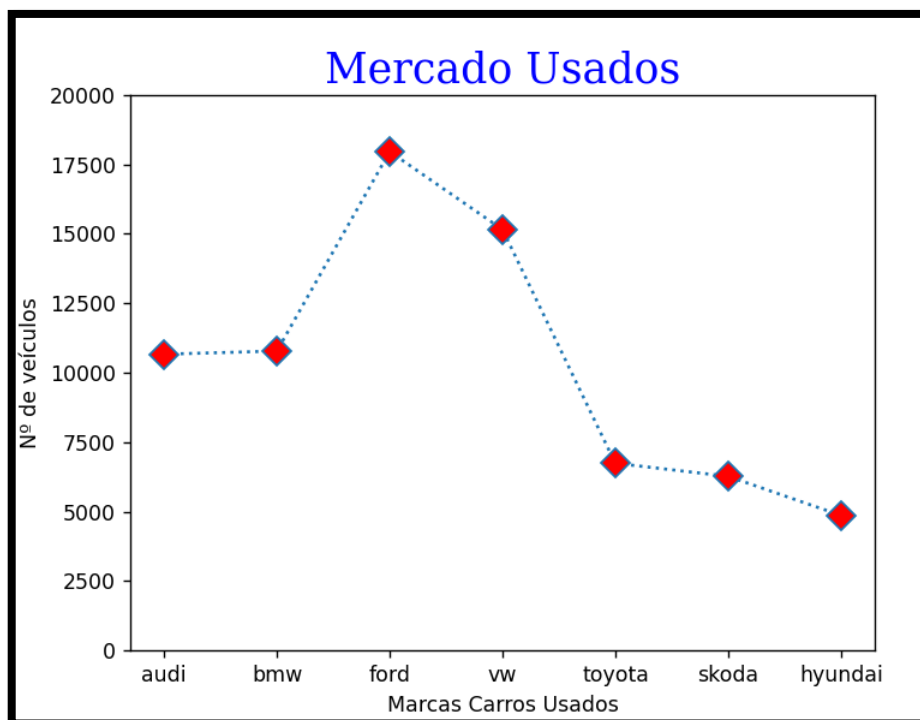
COURSE UNIT

worksheet 08 – Data Visualization: Matplotlib Library

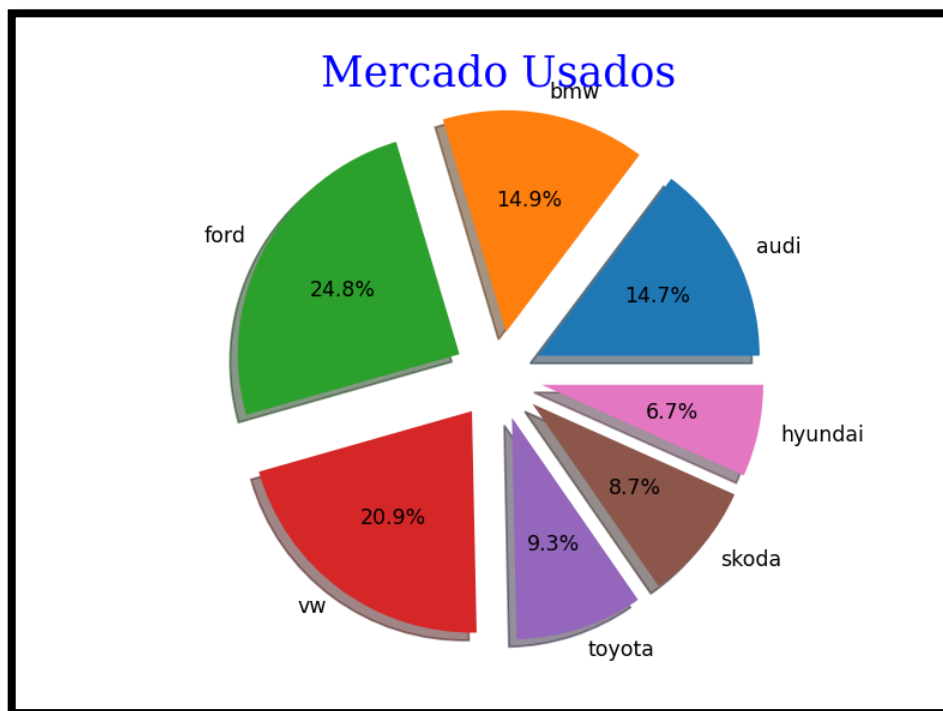
Open the file **carros_usados.csv** file which is a *dataset* of data relating to the sale of cars from a German company (made available for free, like many other datasets, through the <https://www.kaggle.com/datasets> platform).

1. Get the number of used cars per brand.

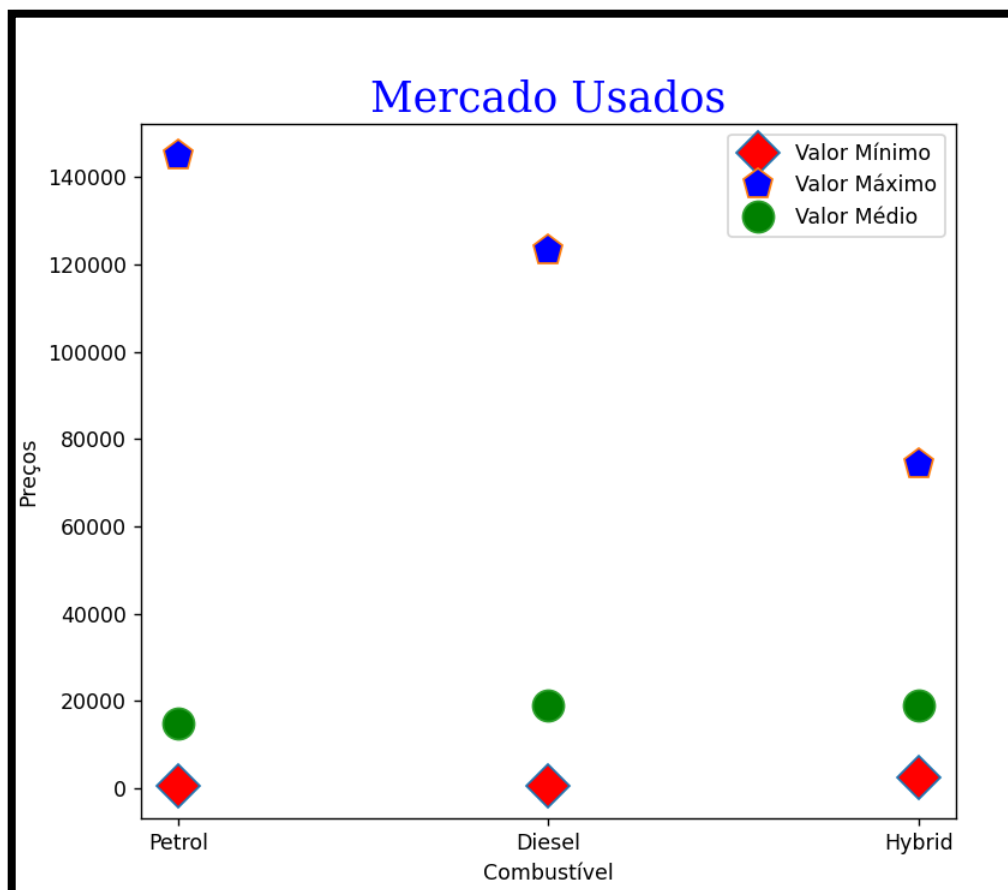
Print the data on the console in the form of a table and produce a graph similar to the one shown below.



2. Produce a chart, based on the file data, with the percentage of each brand.

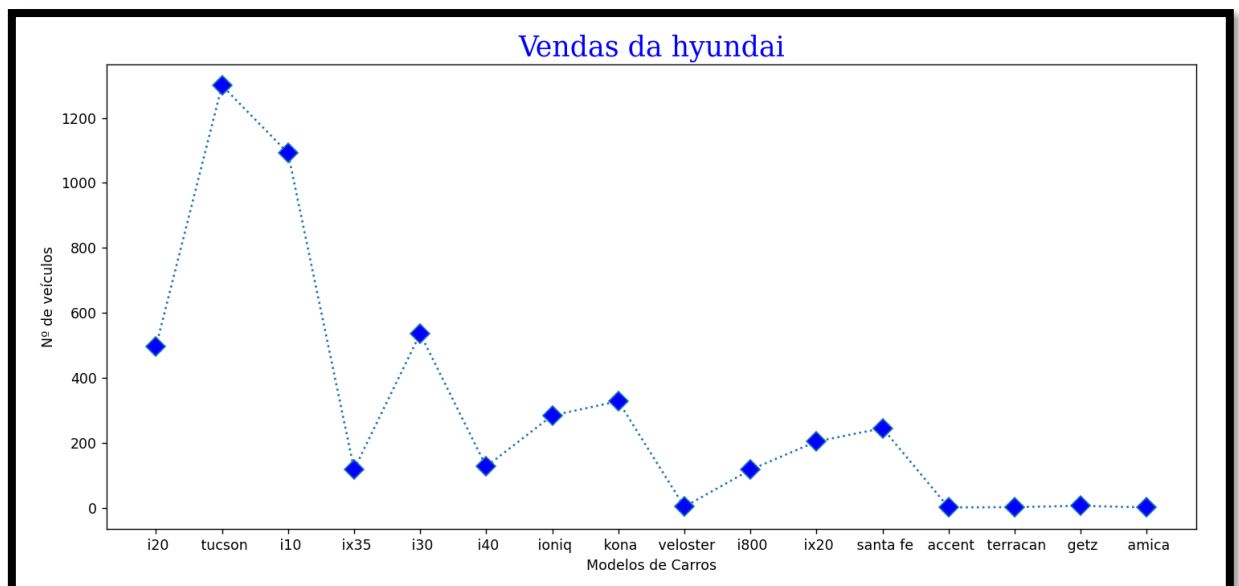
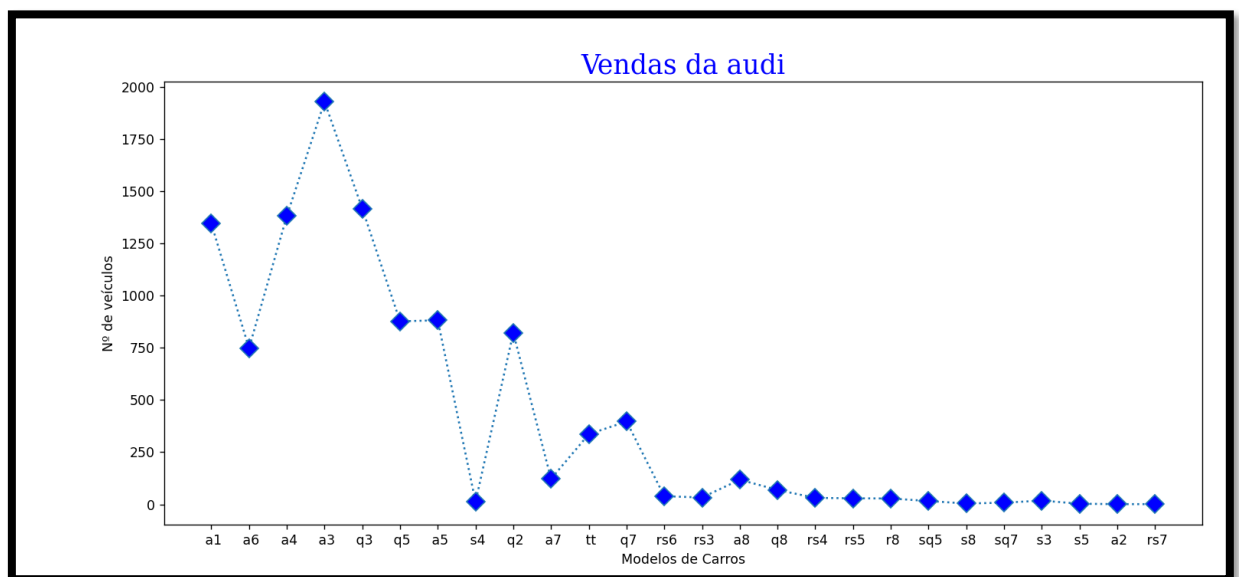


3. Reder a chat with the maximum, minimum and average price, per Fuel Type.



4. Elaborate a dashboard with a set of charts that represent the car sales by brand / model, such as represented below. You should have a chart by brand, through calling a function with different parameters.

```
1 chartPlot4('audi')
2 chartPlot4('hyundai')
```



5. Create a **bar chart** with the number of sales (number of cars) for a specific brand, indicated by user. The chart should distinguish the number of cars by fuel type.

6. Create a pie chart, with percentual data, based on Fuel Type. That is, the percentage of gasoline, diesel, and hybrid cars sold.